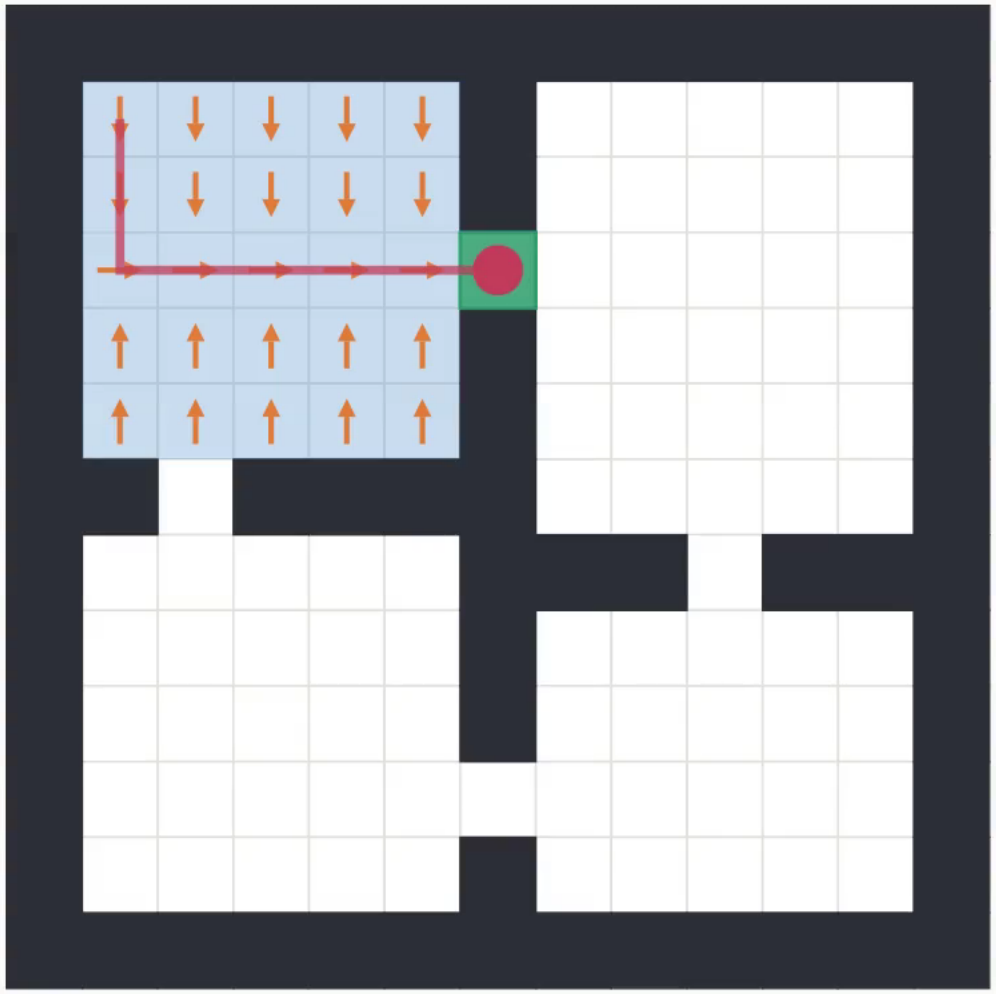
Option Keyboard

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RLAI Seminars
Nov 13, 2026

What is an option?



$$o = \langle \mathcal{I}, \pi, \beta \rangle$$

"go to the hallway"

$\mathcal{I}$ where it can start

π how it acts

β where it stops

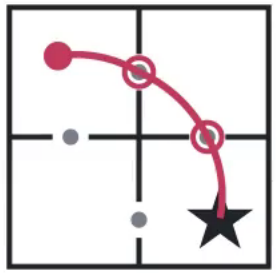
1 decisions

7 steps

one decision, many steps

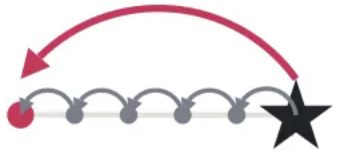
Why should an agent use options?

1



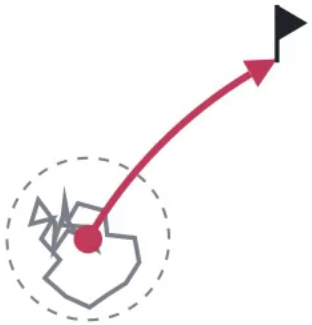
Efficient planning

2



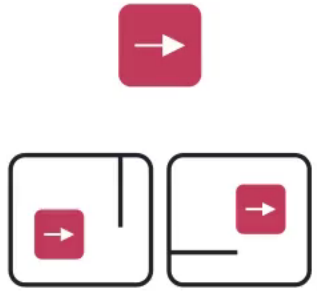
Efficient credit assignment

3



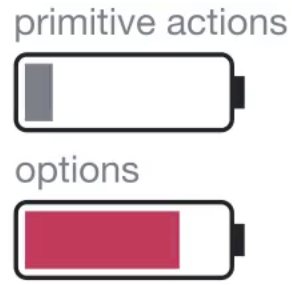
Directed exploration

4



Transfer & reusability

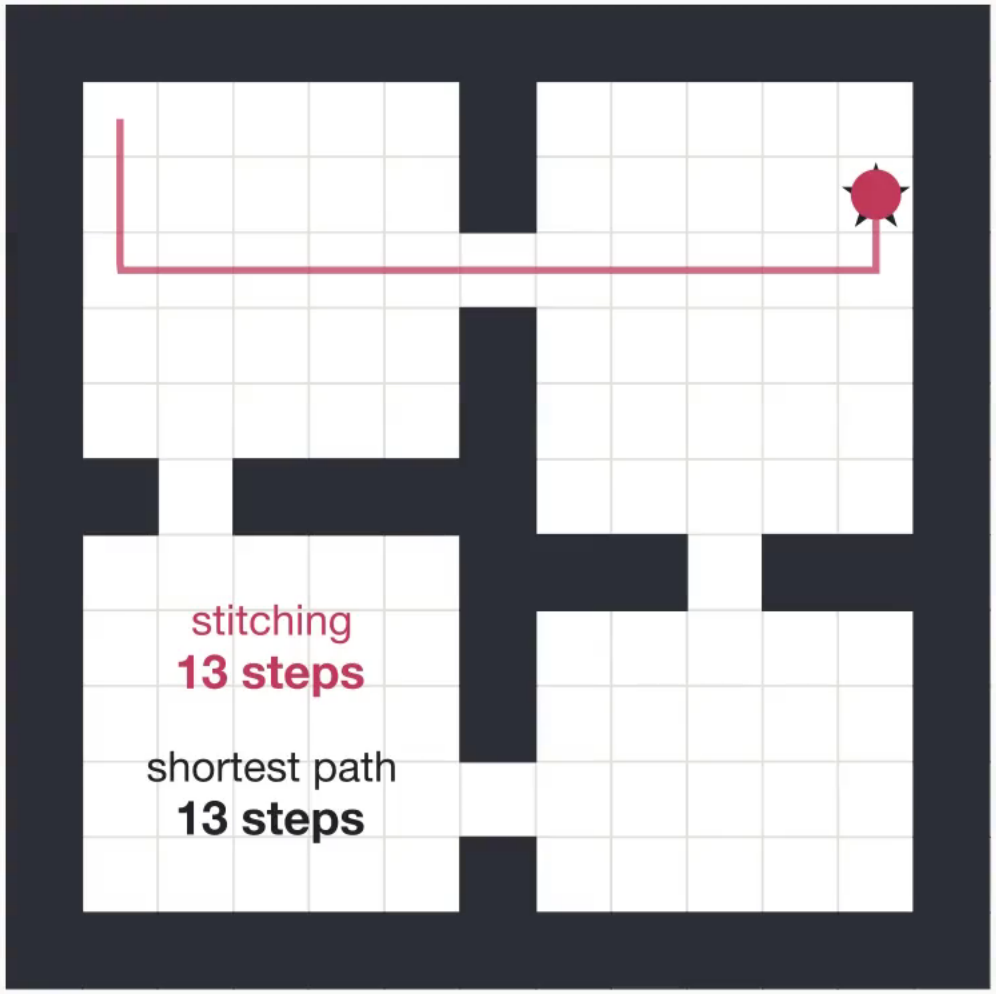
5

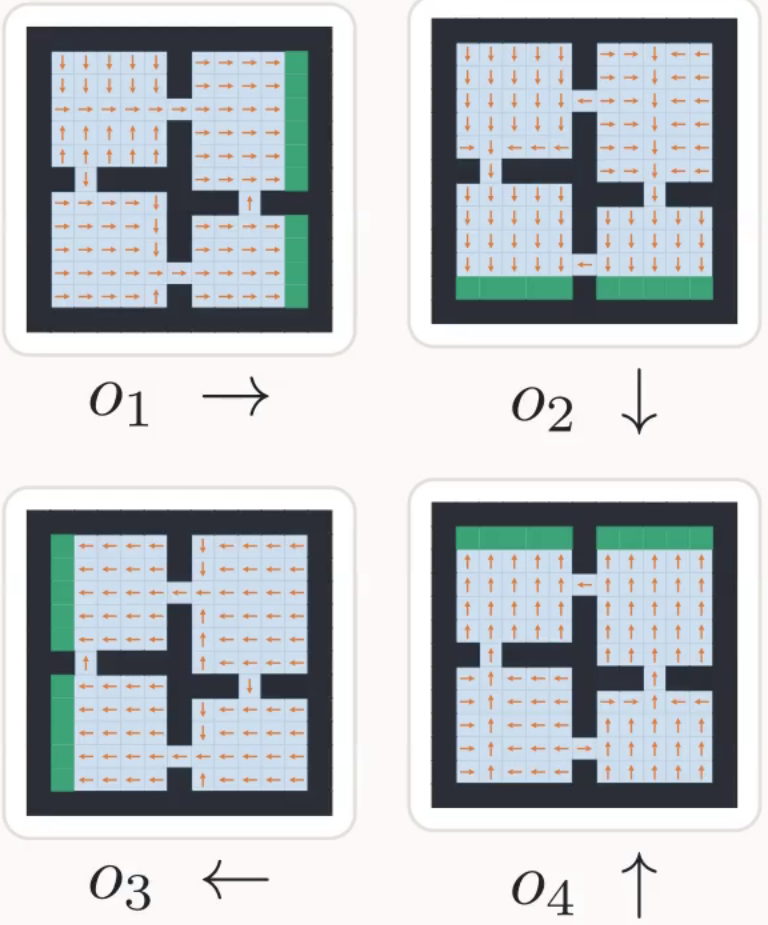


Lower energy expenditure

Task A: stitch options to reach ★

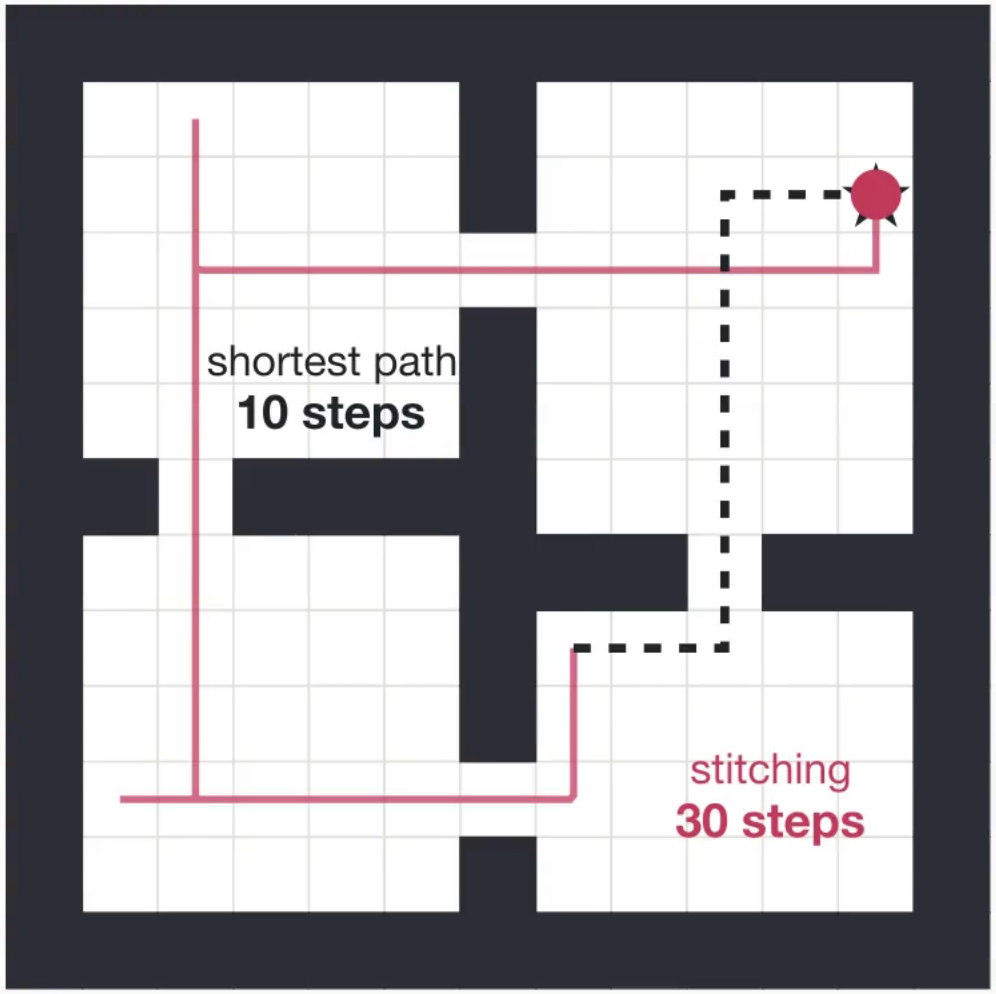
here stitching finds the shortest path

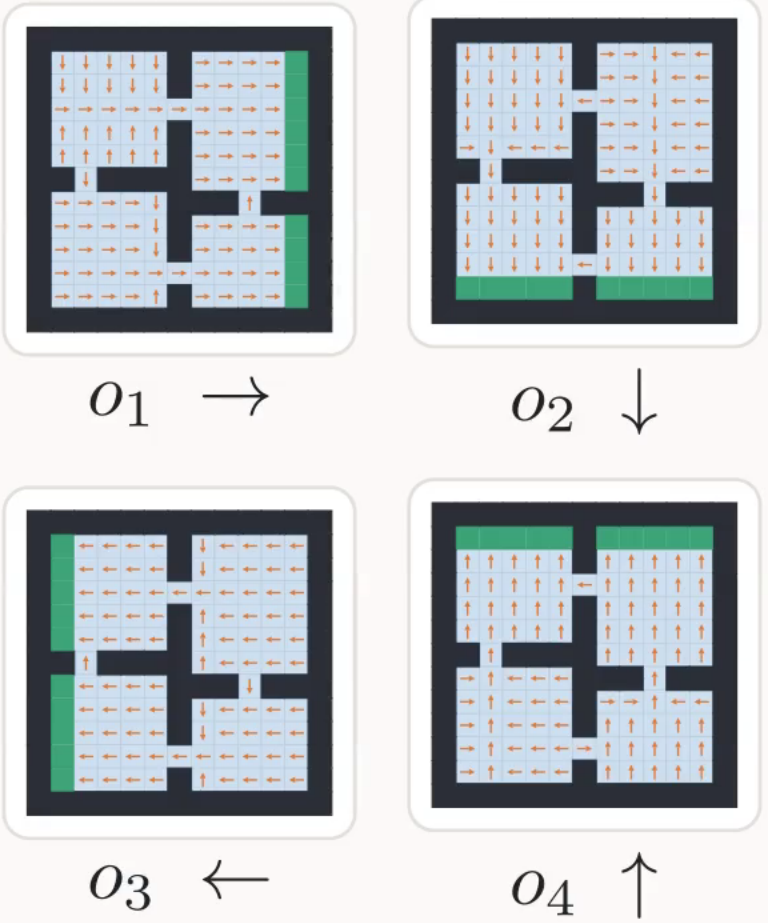




General, not task-specific

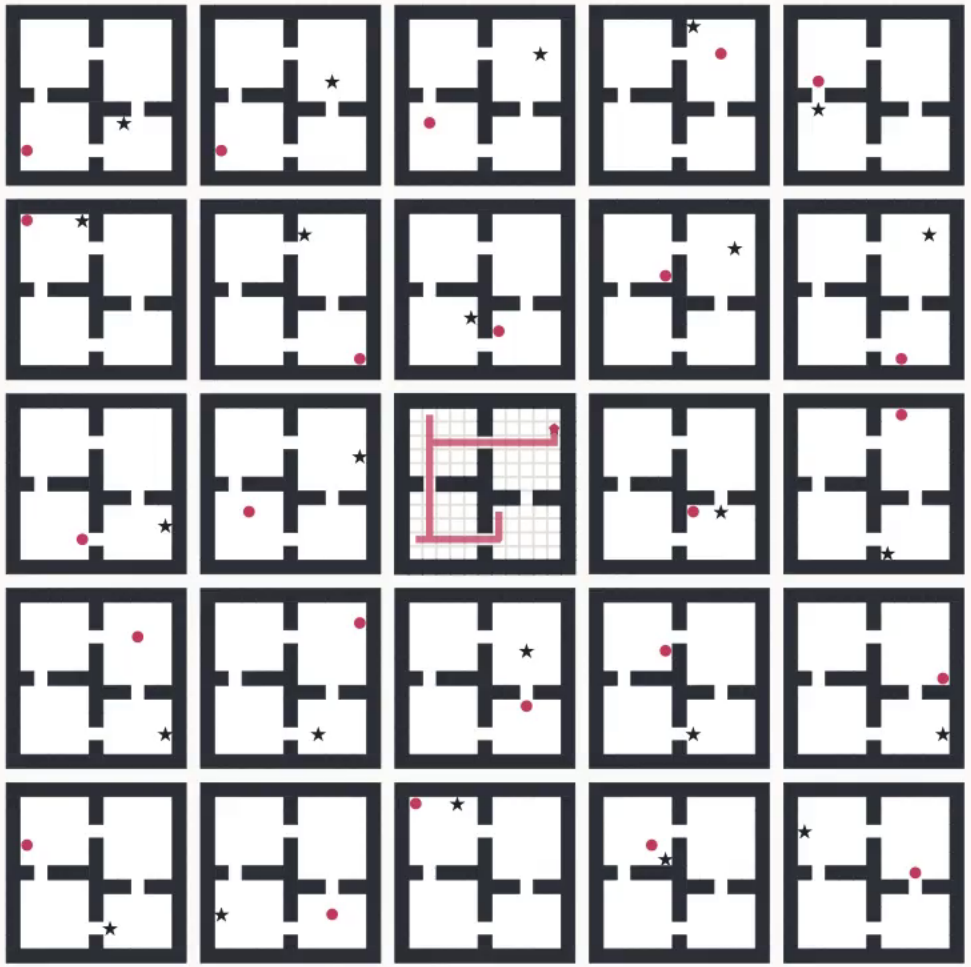
stitching still reaches ★, but the long way round



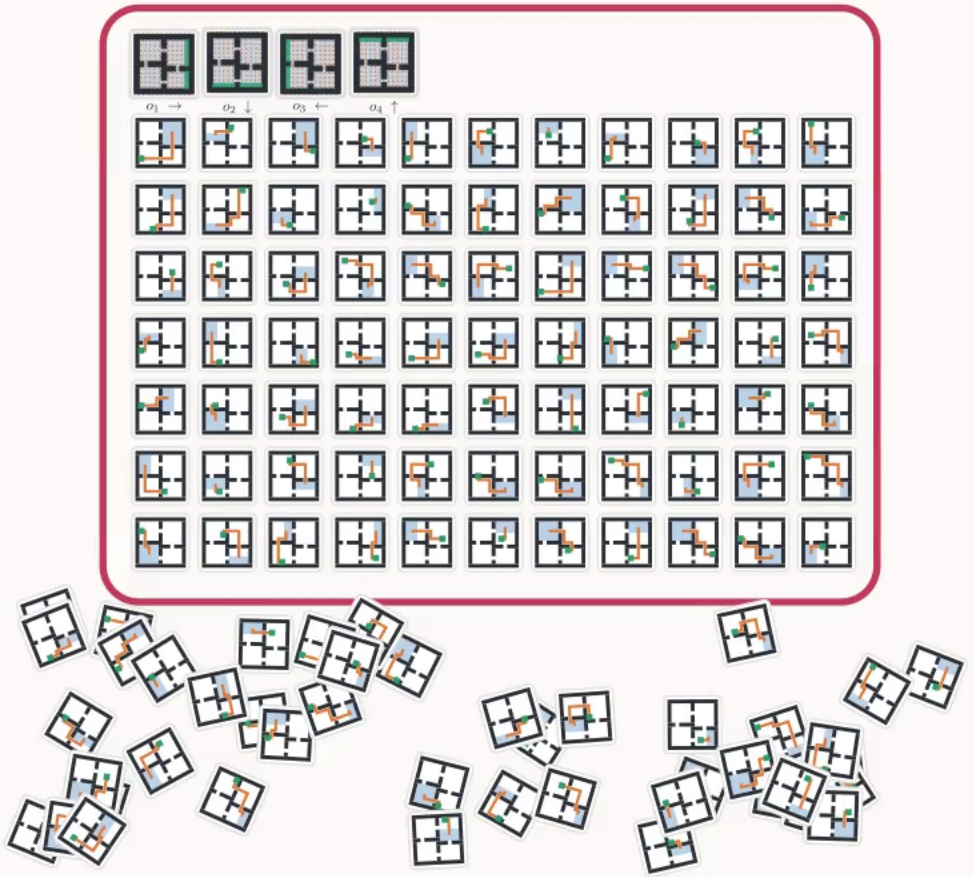


The world is big

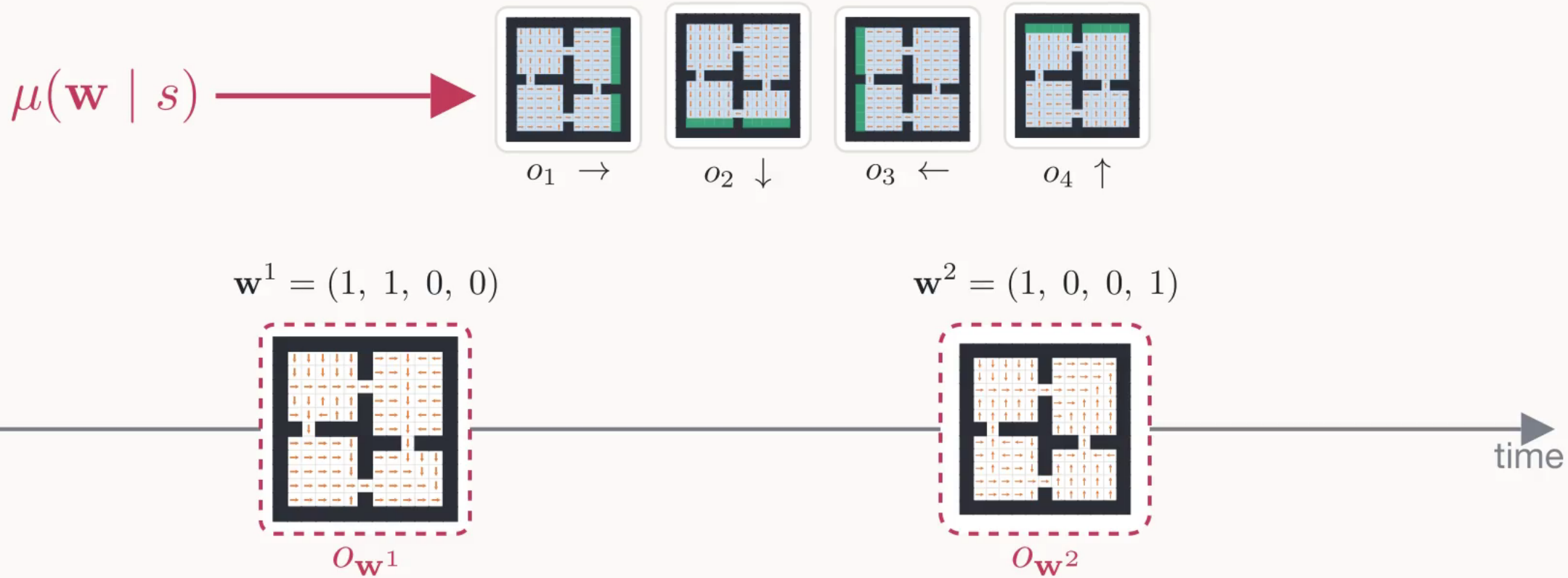
can't store a policy for every task or sub-task



option library # options: ∞



So how do we use a set of options?

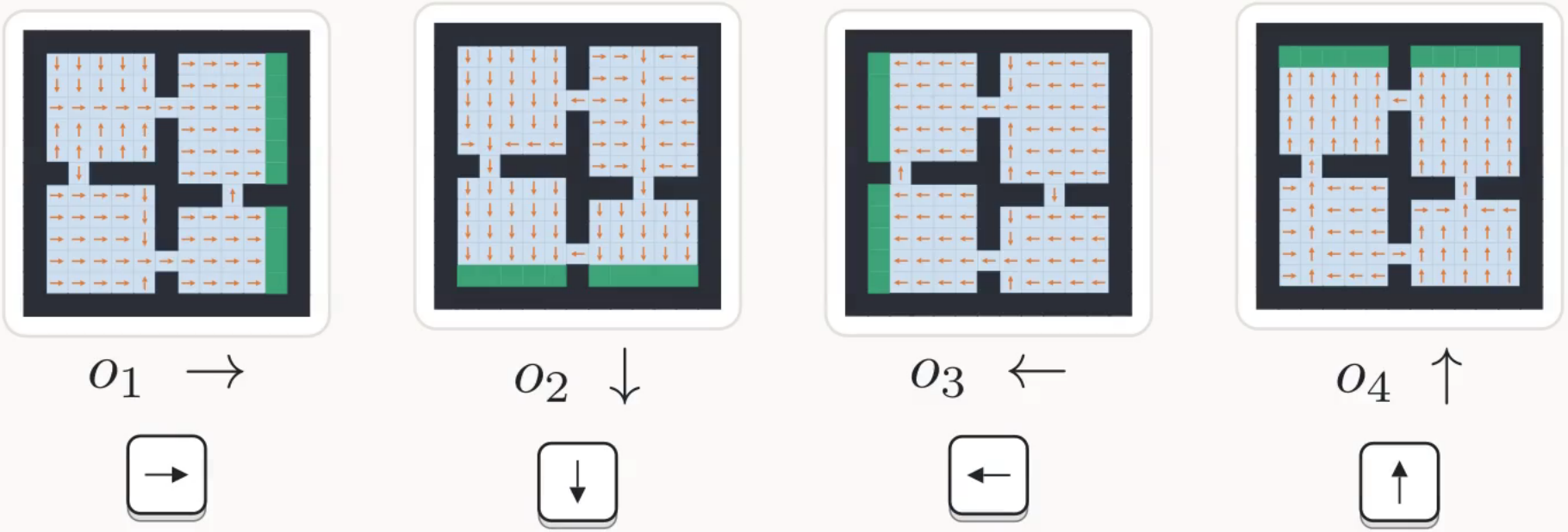


3 Both: the Option Keyboard

learn a policy over chords; each chord makes a new option

The Option Keyboard

the four options from slide 3, one per direction

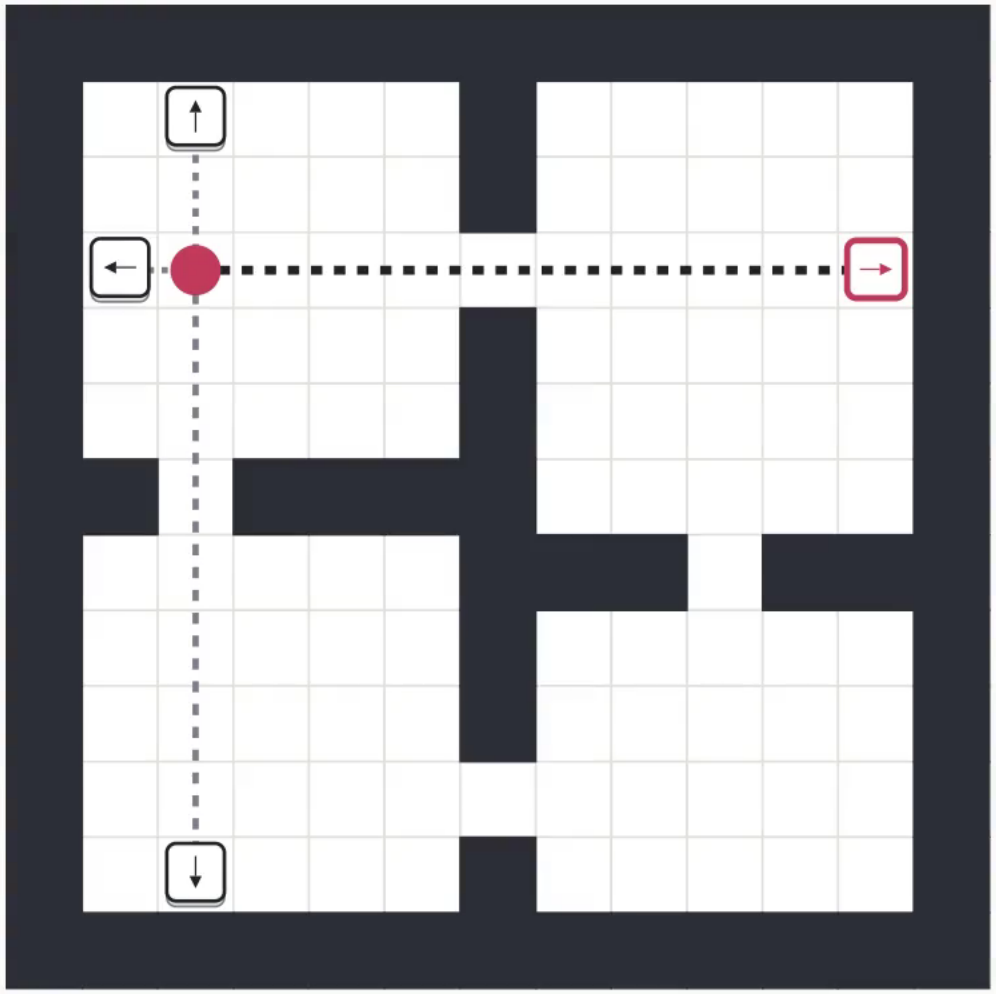


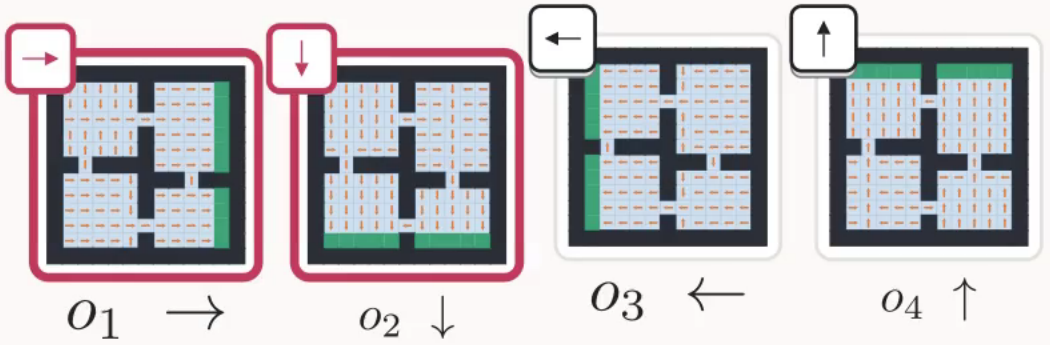
Initiation set: the whole state space

each option can be started in any state, like a key you can press at any time

How the keyboard plays a chord

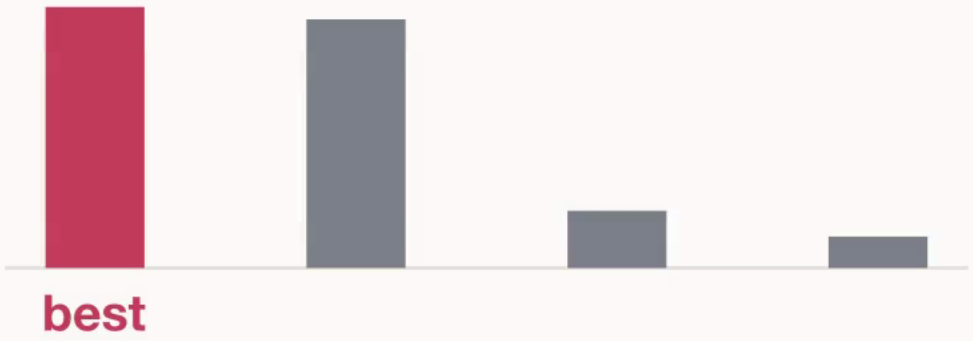
no new learning: each key already knows where it would take you





$\mathbf{w} = \quad 1 \quad 1 \quad 0 \quad 0$

score = how far east + south it gets you (sooner counts more)



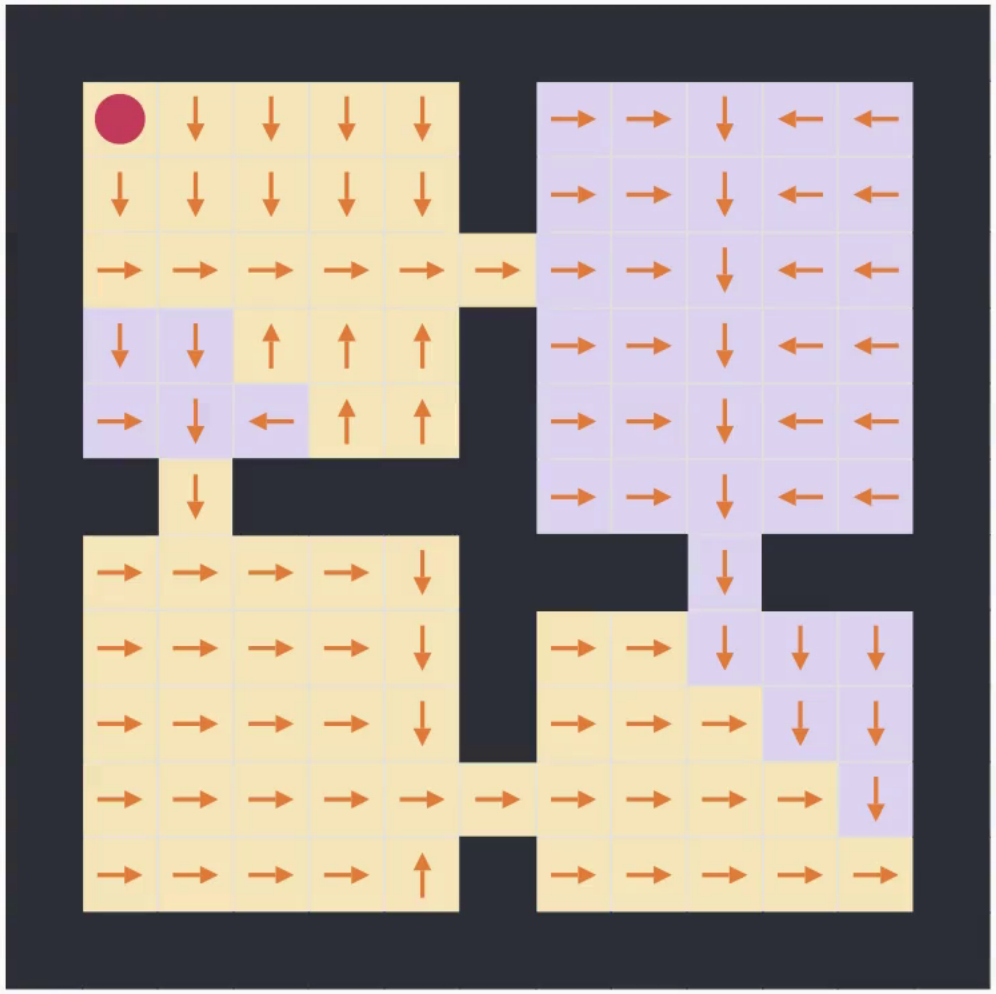
① imagine each key's future, score it (GPE)

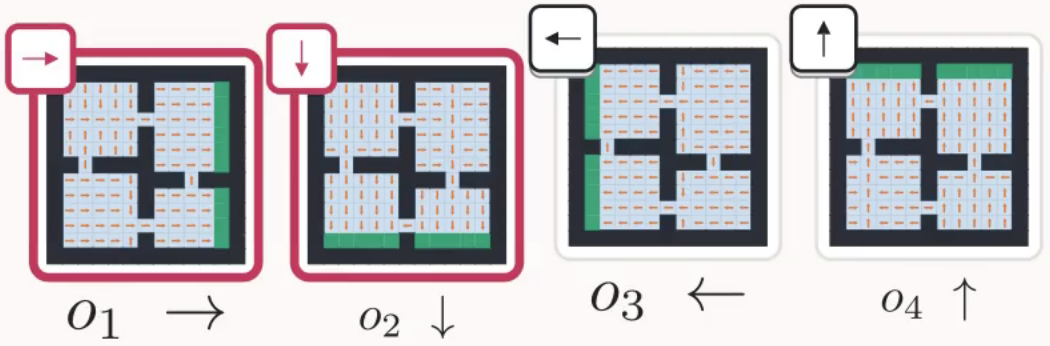
② take the best key's step, then ask again (GPI)

GPE: generalized policy evaluation · GPI: generalized policy improvement

Do that in every cell: a new option

o_1's moves in some cells, o_2's in the rest: nobody trained this





$\mathbf{w} = \quad 1 \quad 1 \quad 0 \quad 0$

o_1 followed here

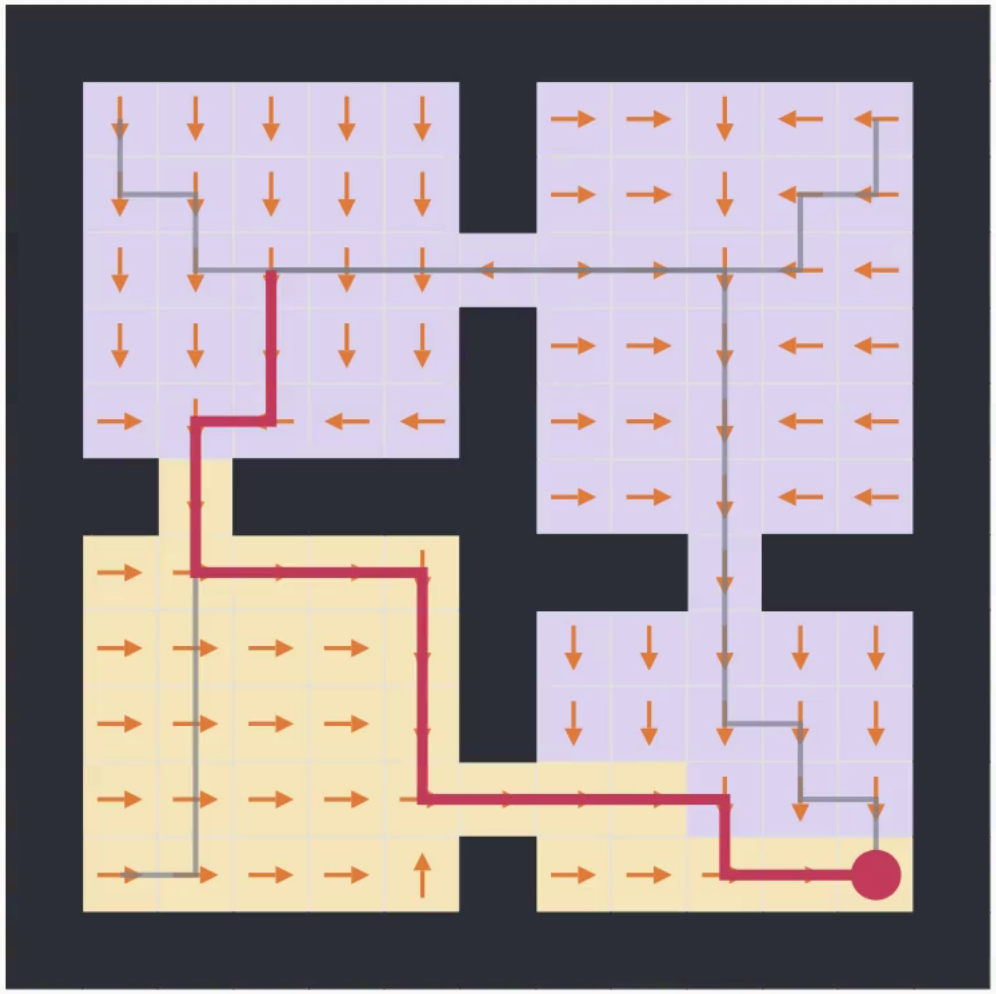
o_2 followed here

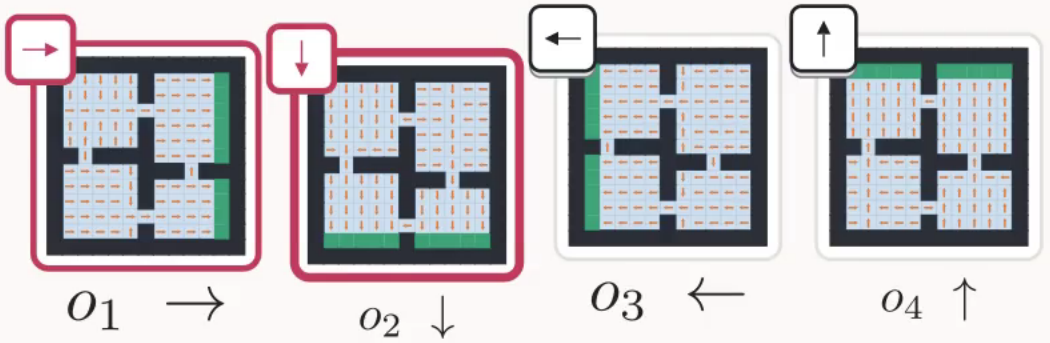
$o_1 + o_2 = o_{\mathbf{w}}$ **a new option**

built from the keys alone: no extra learning

Change the chord, change the behavior

same four keys, same start: only how hard each key is pressed





$\mathbf{w} =$ 0.5 1 0 0

$(1, 1, 0, 0)$ **east + south: to the bottom-right**

$(1, 0, 0, 1)$ **east + north: to the top-right**

$(0, 1, 1, 0)$ **south + west: to the bottom-left**

$(0, 0, 1, 1)$ **west + north: to the top-left**

$(0.5, 1, 0, 0)$ **press south harder: a new route**

every chord is a different option

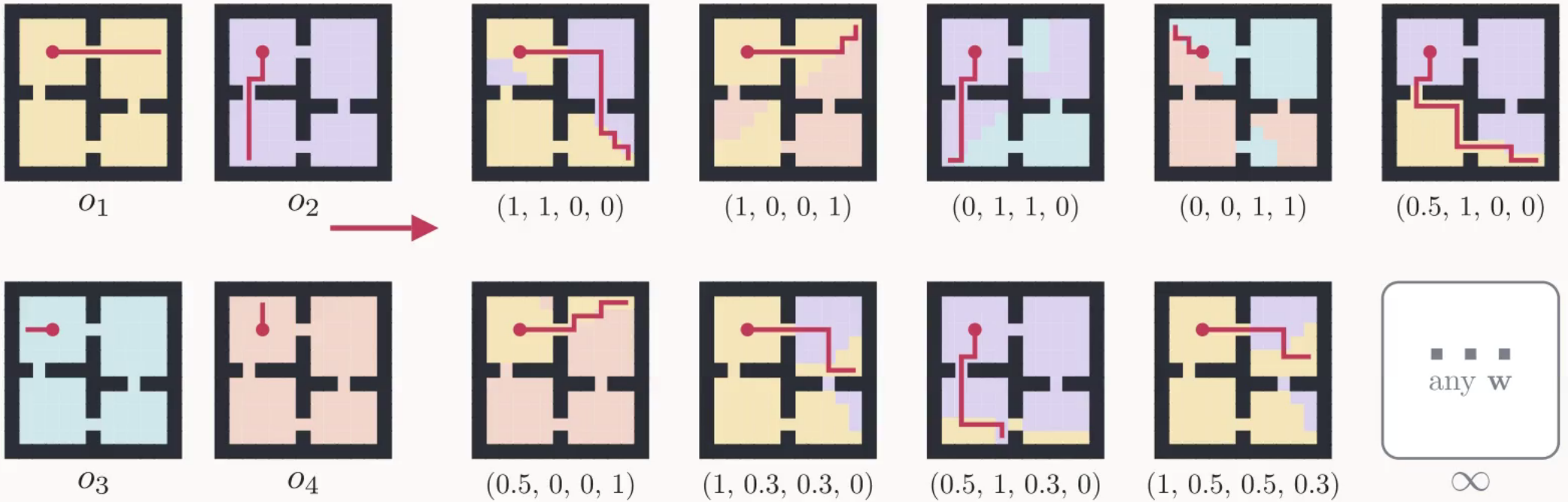
all from the same four keys, nothing retrained

4 options in, a whole space of behaviors out

each chord is its own option, built on the fly from the same four keys

what we had: 4 options

what chords give us

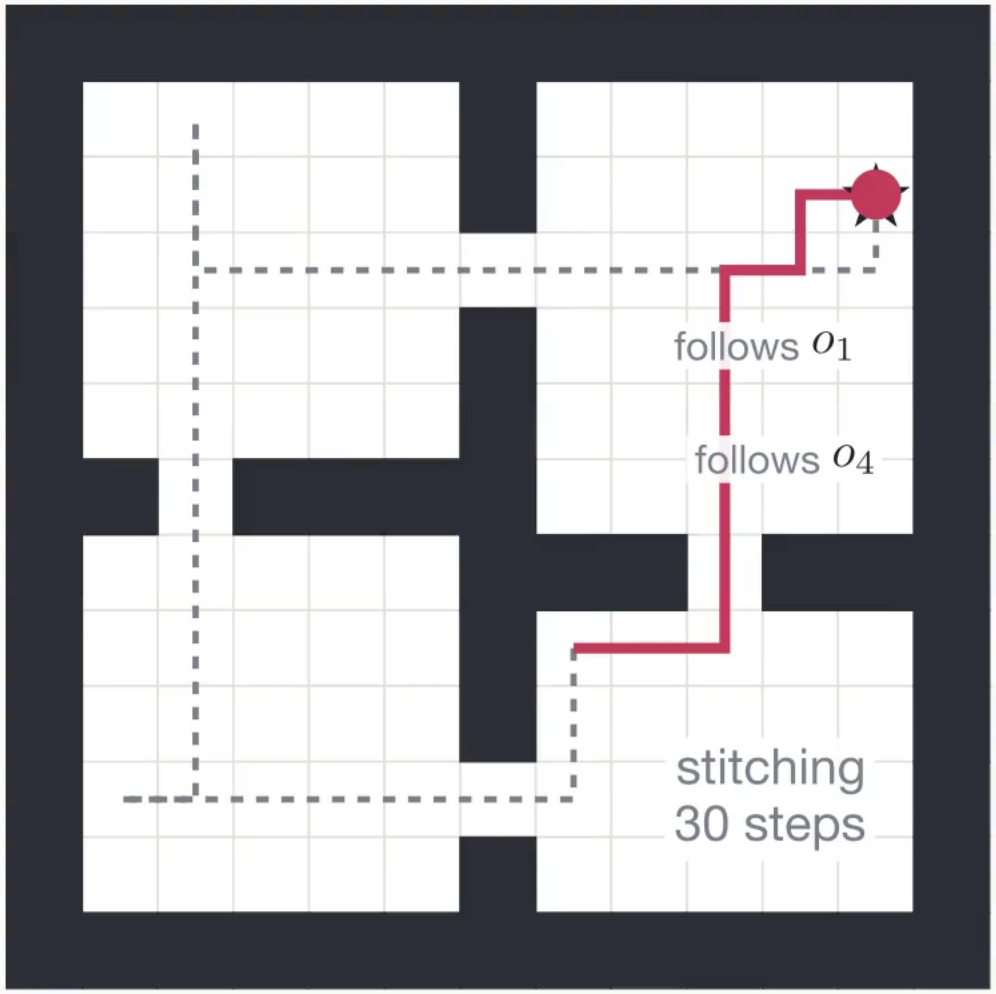


press each key at one of 5 strengths $\{0, 0.25, 0.5, 0.75, 1\}$ $\Rightarrow$ **48 different behaviors, not 4**

…and every weighting in between: combinations, not new training

Task B: one chord does it

the start where stitching our options took the long way round



state

Player $\mu(\mathbf{w} \mid s)$

picks which chord to play, now and then

learned with RL on the task's reward

chord $\mathbf{w}^1 = (1,\ 0,\ 0,\ 1)$

Keyboard

picks every action

GPE + GPI, no learning

o_1 o_2 o_3 o_4

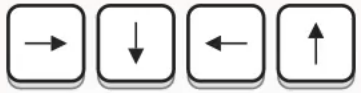
action, every step

1	10
decisions	steps

stitching	$o_3 \to o_4 \to o_1 \to o_4$	4 decisions · 30 steps
keyboard	$o_1 + o_4$	1 decision · 10 steps

the shortest possible route is 10 steps

The Option Keyboard



keys the options you already have

learned once, reused



chords press keys together → a new option

GPE + GPI: no extra learning

$\mu(\mathbf{w} \mid s)$

player which chord, when

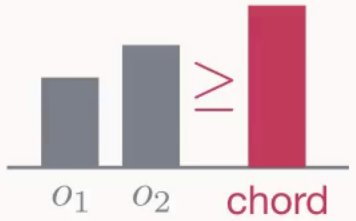
learned for each task (RL)

combine options with chords + stitch chords with a player

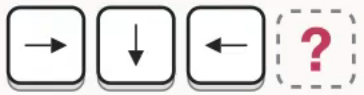
Conclusions & tradeoffs



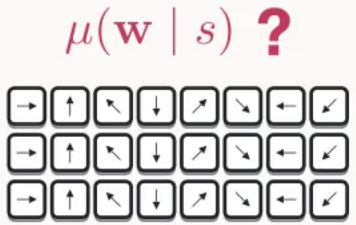
chords build new options from the old ones, with no extra learning



each new behavior is at least as good as any of the keys it was built from



but it can only play what its keys allow: it needs a good set of options to start



more keys mean many more chords, so the player gets harder to learn